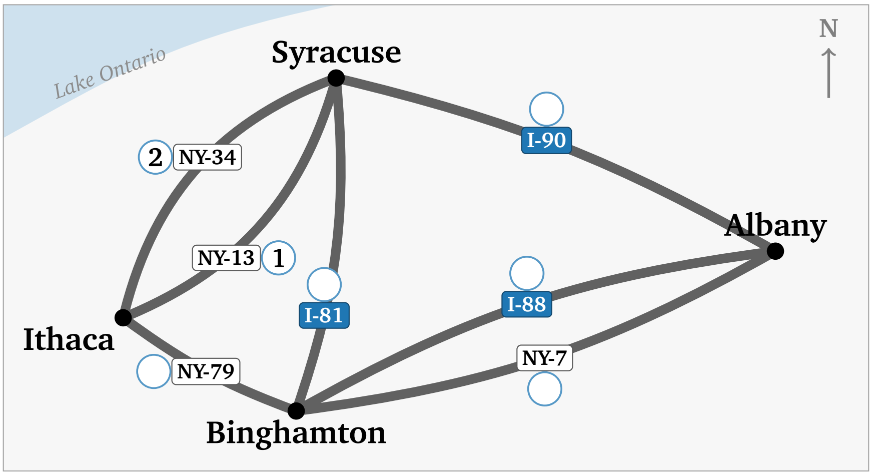
Drive Every Highway Once, Then Teach a Computer to Do It

Do this before the lecture, in pencil. Nothing here needs a word you have not been given on the sheet.

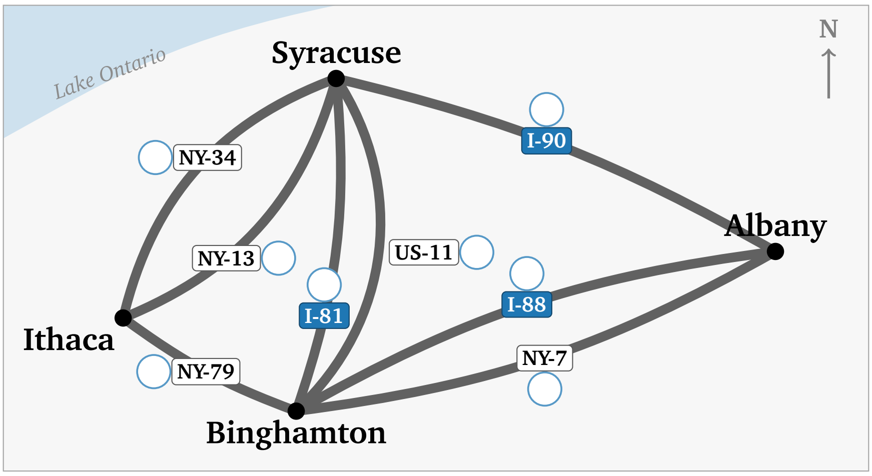
Part 1 — Seven highways

Seven highways join four Upstate New York cities: **I**thaca, **S**yracuse, **B**inghamton, **A**lbany.

Question 1(a): Drive every highway exactly once, without lifting your pencil. Start and finish anywhere; cities may repeat, highways may not. Write in each circle when you drive that road — 1 first, then 2, on to 7. Two are filled in: you start at Ithaca, take NY-13, then NY-34 straight back.



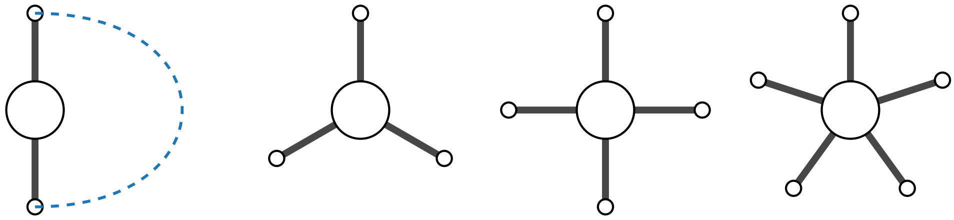
Question 1(b): The same four cities, and an eighth highway: **US-11**, which really does run alongside I-81. Number your route again.



Can you drive every highway without lifting your pencil? ____________________

Question 2: One city, on its own. Arriving uses up one highway, leaving uses up another — so highways get used in **pairs**.

Join two road-ends with a curved line to mean “in here, out there”. Pair up as far as you can. The first is done.



2 roads 3 roads 4 roads 5 roads

	2 roads	**3 roads**	**4 roads**	**5 roads**
pairs you drew	1			
roads left over	0			

(a) Which ones can you drive straight *through*? ______________________

(b) What do those numbers share? __________________________

(c) A left-over road has no partner, so this city must be the tour’s ____________________

Question 3: Count the highways at each city. Ithaca is done.

	I	**S**	**B**	**A**	**how many have a left-over?**
seven roads	3				
with US-11 too					

One start, one finish. So at most _____ cities may have a left-over road.

Seven roads: ________________ Eight roads: ________________

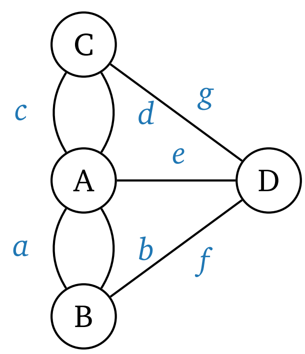
No route with eight because _____________________________________

Question 4: The same puzzle, 1736. Königsberg: four pieces of land, seven bridges. Two pairs join the same land; both of a pair must be crossed.

Bridges: A ____ B ____ C ____ D ____

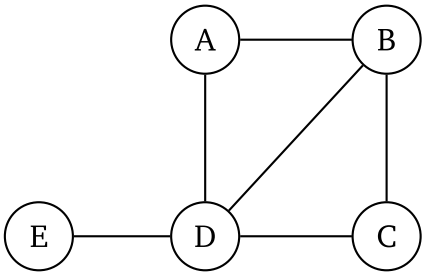
Is the walk possible? _______________

Destroying one bridge is enough. Which ones work? _____________



Part 2 — Three kinds of route

A different network: five bus stops.



Question 5(a): Fill in the table. The last column takes one of these three names. First row done.

Repeats nothing: **path**. Never repeats a line: **trail**. Repeats both: **walk**.

route	repeats a line?	repeats a stop?	name
A B D B C	yes (B–D twice)	yes (B)	
A B D C B			
A B C D E			

Question 5(b): A route from A to E that repeats a stop but no line:

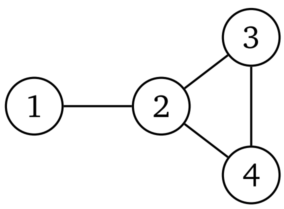
A _______ _______ _______ _______ E

Question 5(c): Back to the highway map. Your drive in 1(a) used no road twice, but it did come back to a city it had already visited. It is a ________________.

Question 5(d): It could not have been a path. A path of 7 roads would have to touch ________ different cities, and the highway map has only ________.

Part 3 — Predictions for the notebook

Part 4 sends you to a notebook that will run this network for you. Work the answers out on paper first; the notebook will check them.



Question 6(a): Write **1** in row i, column j if a line joins i and j, else **0**. Row 1 done.

$A =$	1	2	3	4
1	0	1	0	0
2				
3				
4				

Question 6(b): Add up each row, then compare with the picture.

row totals: _____ _____ _____ _____

Each row total is the ______________________

of that node.

Question 6(c): A **2-step route** goes along one line, then along one more. From node 1 to node 3, $1 \to 2 \to 3$ is one such route. Using the picture, count the 2-step routes:

from 1 to 3: how many? ______ write one out: $1 \to$ _____ $\to 3$

from 1 to 2: how many? ______ why so few? ______________________

from node 2 back to node 2: how many? ______

Question 6(d): The notebook at `go.skojaku.com/m01lab` will square your grid for you. Before it does, say what you expect to find in A^2.

row 1, column 3: _______ row 1, column 2: _______

the diagonal: ______________________________________

Part 4 — The lab, on your own

Do this one by yourself, with this sheet beside you. The notebook takes the same four cities and seven highways you have been drawing on, and asks you to write them down in a form a computer can read.

`go.skojaku.com/m01lab`